

# HKU Unofficial Note (Introductory EM)

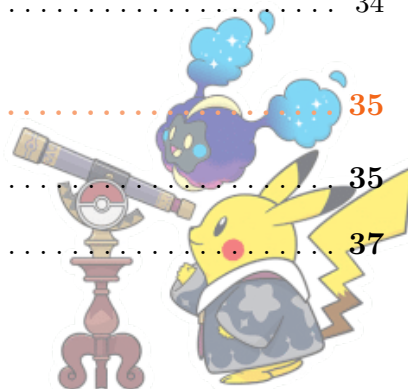
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# 1 Review and Further Studies of DSE Level Concepts

## 1.1 Lorentz's Force

The **Lorentz force** describes the total force experienced by a charged particle moving in electric and magnetic fields. It is given by

$$\mathbf{F} = q (\mathbf{E} + \mathbf{v} \times \mathbf{B})$$

where

- $q$  is the charge of the particle,
- $\mathbf{E}$  is the electric field, contributing the electric force,
- $\mathbf{v}$  is the velocity of the particle, and
- $\mathbf{B}$  is the magnetic field, contributing the magnetic force.

## 1.2 Coulomb's Law and Continuous Charge Distribution

Coulomb's law describes the force between two point charges. If two point charges  $q_1$  and  $q_2$  are separated by a distance  $r$ , the electrostatic force  $\mathbf{F}$  between them is given by

$$\mathbf{F} = k_e \frac{q_1 q_2}{r^2} \mathbf{r}$$

where

- $k_e = \frac{1}{4\pi\epsilon_0} \approx 8.9875 \times 10^9 \text{ N m}^2/\text{C}^2$  is Coulomb's constant,
- $\epsilon_0$  is the vacuum permittivity, and
- $\mathbf{r}$  is the unit vector pointing from  $q_1$  to  $q_2$ .

For extended objects, charges are often distributed continuously rather than concentrated at points. In such cases, we define a **charge density**:

- Linear charge density:  $\lambda = \frac{dq}{dl}$ , for a line of charge.
- Surface charge density:  $\sigma = \frac{dq}{dA}$ , for a surface charge.





- Volume charge density:  $\rho = \frac{dq}{dV}$ , for a volume charge.

The infinitesimal force  $d\mathbf{F}$  exerted on a test charge  $q_0$  by a small charge element  $dq$  is

$$d\mathbf{F} = k_e \frac{q_0 dq}{r^2} \mathbf{r}$$

To find the total force, integrate over the entire distribution

$$\mathbf{F} = k_e q_0 \int \frac{dq}{r^2} \mathbf{r}$$

### 1.3 Electric Potential Energy and Electric Potential

The **electric potential energy** of a system describes the energy stored due to the positions of charges relative to each other. For a point charge  $q$  in the electric field created by another point charge  $Q$ , the electric potential energy  $U$  is given by

$$U = k_e \frac{Qq}{r}$$

where

- $Q$  and  $q$  are the magnitudes of the charges, and
- $r$  is the distance between the charges.

The work  $W$  done by an external force to move a charge  $q$  from point  $A$  to point  $B$  in an electric field is related to the change in electric potential energy

$$W_{AB} = U_B - U_A$$

The **electric potential** at a point in space is defined as the electric potential energy per unit charge:

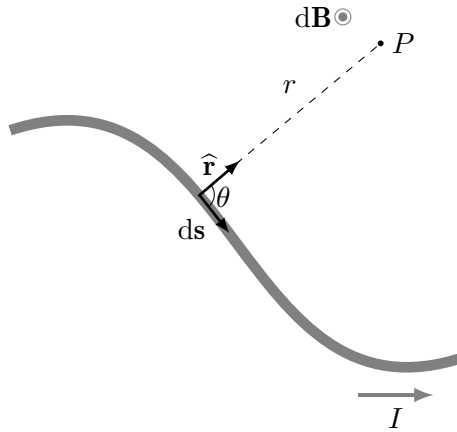
$$V = \frac{U}{q}$$

### 1.4 Biot–Savart law

The **Biot–Savart law** gives the magnetic field produced by a small segment of current-carrying conductor. It is written as

$$d\mathbf{B} = \frac{\mu_0}{4\pi} \frac{I d\mathbf{l} \times \mathbf{r}}{r^2}$$





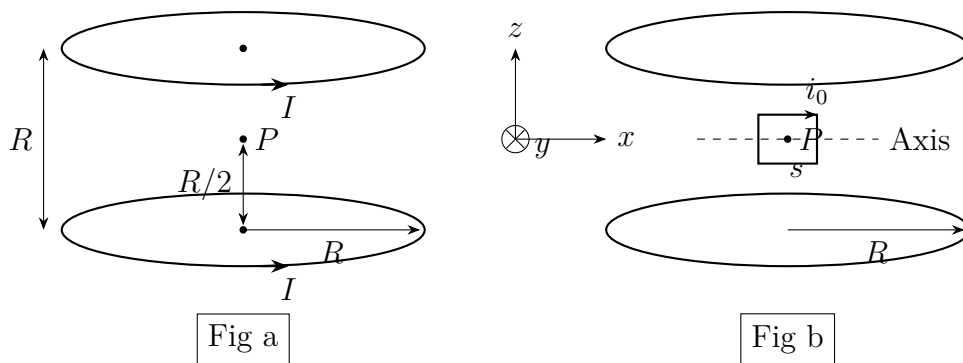
Hence, for a circular loop of radius  $R$  carrying current  $I$ , the magnetic field at the center of the loop is

$$B = \frac{\mu_0 I}{2R}$$

The Biot-Savart law is consistent with Ampère's law, which states that

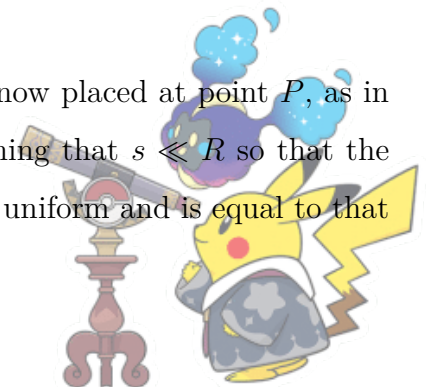
$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{enc}}$$

**Example.** Two circular loops of wire lying parallel to the  $xy$ -plane each has a radius  $R$  and each carries a current  $I_0$ . The two loops of wire are aligned with their centers at a distance  $R$  from each other, as shown in Fig a. (The loops are drawn so that the lower side is closer to the observer.)



- (a) Using Biot-Savart law, determine the magnetic field  $B_{\text{tot}}$  (magnitude and direction) due to the two current loops at point  $P$ , which is half the distance between the line joining the centers of the two loops.

A small square loop of current  $i_0$  in which each side has a length  $s$  is now placed at point  $P$ , as in Fig 4b. The loop is allowed to rotate about an axis as shown. Assuming that  $s \ll R$  so that the magnetic field at the position of the square loop can be assumed to be uniform and is equal to that at  $P$ .



- (b) The plane of the square loop is tilted from the axis as shown and makes an angle of  $30^\circ$  with the  $xy$ -plane. Determine the net force and net torque experienced by the small square loop, and describe the motion of the loop, including its final state. Express your answer in terms of  $B_{\text{tot}}$ ,  $i_0$ ,  $s$ , and other constants.
- (c) What is the work done by the external force on the small square loop in moving it from the initial to the final position?
- (a) The magnetic field on the axis of a circular loop of radius  $R$  at a distance  $z$  from its center is given by

$$B = \frac{\mu_0 I R^2}{2(R^2 + z^2)^{3/2}}$$

**Proof.** The Biot-Savart Law states

$$d\mathbf{B} = \frac{\mu_0}{4\pi} \frac{I d\mathbf{l} \times \mathbf{r}}{r^2}$$

The magnitude of this differential field is

$$dB = \frac{\mu_0 I}{4\pi} \frac{dl \sin(90^\circ)}{r^2} = \frac{\mu_0 I}{4\pi} \frac{dl}{R^2 + z^2}$$

For every current element  $d\mathbf{l}$  on one side of the loop, there is an identical element on the opposite side. Their radial components cancel each other out due to symmetry. The components along the  $z$ -axis add up.

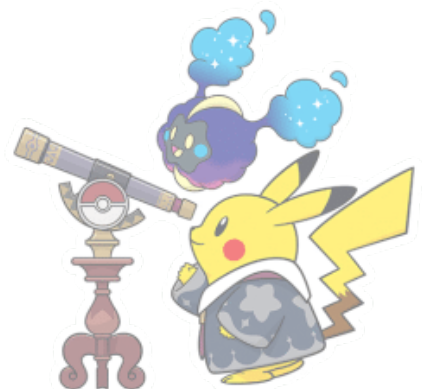
From the geometry of the triangles, the axial component is  $dB_z = dB \cos(\phi)$ , where  $\phi$  is the angle  $d\mathbf{B}$  makes with the  $z$ -axis. Note that by similar triangles,  $\cos(\phi) = \frac{R}{\sqrt{R^2 + z^2}}$ . To find the total field  $B$ , we integrate the  $z$ -components around the entire loop:

$$B = \int dB_z = \int dB \cos(\phi)$$

Substitute the expressions for  $dB$  and  $\cos(\phi)$ :

$$B = \int \left( \frac{\mu_0 I}{4\pi} \frac{dl}{R^2 + z^2} \right) \left( \frac{R}{\sqrt{R^2 + z^2}} \right)$$

$$B = \frac{\mu_0 I R}{4\pi(R^2 + z^2)^{3/2}} \int dl$$



The integral  $\int dl$  is simply the circumference of the loop, which is  $2\pi R$ .

$$B = \frac{\mu_0 I R}{4\pi(R^2 + z^2)^{3/2}} \cdot (2\pi R) = \frac{\mu_0 I R^2}{2(R^2 + z^2)^{3/2}}$$

Point  $P$  is located at  $z = R/2$  from the center of the bottom loop and  $z = -R/2$  from the center of the top loop. As the currents flow in the same direction, the fields from both loops add constructively in the  $+z$  direction.

$$B_{\text{tot}} = B_{\text{bottom}} + B_{\text{top}} = 2 \times \frac{\mu_0 I_0 R^2}{2(R^2 + (R/2)^2)^{3/2}}$$

$$B_{\text{tot}} = \frac{\mu_0 I_0 R^2}{(R^2 + \frac{R^2}{4})^{3/2}} = \frac{\mu_0 I_0 R^2}{(\frac{5}{4}R^2)^{3/2}} = \frac{\mu_0 I_0 R^2}{\frac{5\sqrt{5}}{8}R^3}$$

$$\mathbf{B}_{\text{tot}} = \frac{8\mu_0 I_0}{5\sqrt{5}R} \mathbf{k}$$

(b) In a uniform magnetic field, the net force on any closed current loop is zero.

$$F_{\text{net}} = 0$$

The torque  $\tau$  on a magnetic dipole is given by  $\tau = \mu \times \mathbf{B}$ . The magnetic moment of the square loop is  $\mu = i_0 \cdot \text{Area} = i_0 s^2$ . Since the loop is tilted  $30^\circ$  from the  $xy$ -plane, its normal vector (the direction of  $\mu$ ) makes an angle  $\theta = 30^\circ$  with the  $z$ -axis (the direction of  $\mathbf{B}_{\text{tot}}$ ).

$$\tau = \mu B_{\text{tot}} \sin(30^\circ) = (i_0 s^2) B_{\text{tot}} \left(\frac{1}{2}\right) = \frac{1}{2} i_0 s^2 B_{\text{tot}}$$

(c) The work done by an external force is equal to the change in the potential energy  $U$  of the magnetic dipole in the field:

$$U = -\mu \cdot \mathbf{B} = -\mu B_{\text{tot}} \cos \theta$$

Initial Potential Energy ( $\theta = 30^\circ$ ):

$$U_i = -i_0 s^2 B_{\text{tot}} \cos(30^\circ) = -i_0 s^2 B_{\text{tot}} \frac{\sqrt{3}}{2}$$

Final Potential Energy ( $\theta = 0^\circ$ ):

$$U_f = -i_0 s^2 B_{\text{tot}} \cos(0^\circ) = -i_0 s^2 B_{\text{tot}}$$



Work done ( $W_{ext}$ ):

$$W_{ext} = \Delta U = U_f - U_i$$

$$W_{ext} = -i_0 s^2 B_{tot} - \left( -i_0 s^2 B_{tot} \frac{\sqrt{3}}{2} \right)$$

$$W_{ext} = i_0 s^2 B_{tot} \left( \frac{\sqrt{3}}{2} - 1 \right)$$

## 2 Electric Dipole Moment

An **electric dipole** consists of two equal and opposite point charges  $+q$  and  $-q$  separated by a distance

**d**. The **electric dipole moment  $\mathbf{p}$**  is defined as

$$\mathbf{p} = q \mathbf{d}$$

where

- $q$  is the magnitude of one of the charges,
- $\mathbf{d}$  is the vector pointing from the negative charge to the positive charge.

A dipole placed in a uniform electric field  $\mathbf{E}$  experiences a torque

$$\boldsymbol{\tau} = \mathbf{p} \times \mathbf{E}.$$

The potential energy of the dipole in the field is

$$U = -\mathbf{p} \cdot \mathbf{E} = -pE \cos \theta$$

where  $\theta$  is the angle between  $\mathbf{p}$  and  $\mathbf{E}$ .

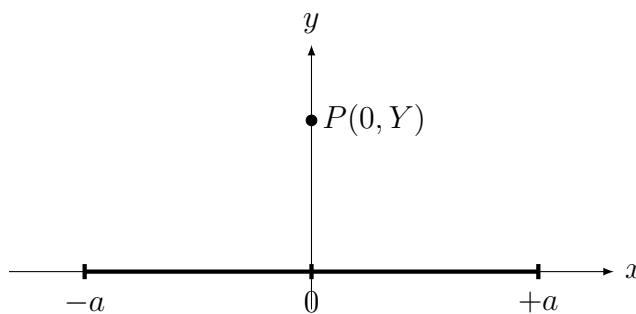
**Example.** Consider a rod of length  $2a$  lying on the  $x$ -axis and centered at the origin. It carries a linear charge density

$$\lambda(x) = \lambda_0 \frac{x}{a}, \quad -a < x < a$$

where  $\lambda_0$  is a positive constant. Hence, the rod is negatively charged for  $x < 0$  and positively charged for  $x > 0$ . Point  $P$  is located at  $(0, Y)$  above the center of the rod.







- (a) Using symmetry arguments, determine the direction of the electric field  $\mathbf{E}_p$  at the point  $P(0, Y)$ .
- (b) Determine the electric potential  $V_p$  at point  $P(0, Y)$  due to the rod. Can you use  $V_p$  to determine  $\mathbf{E}_p$ ? Explain your answer.
- (c) Determine the electric dipole moment of the charged rod.
- (d) Suppose initially the charged rod is held fixed in a region with a uniform external electric field  $\mathbf{E} = E_0(-\mathbf{i} + \mathbf{k})$ . The rod is released from rest and is free to rotate about the origin until it reaches equilibrium. Determine the work done by the external electric force.
- (a) The positive charge at  $+x$  creates an electric field at  $P$  pointing away from itself (up and to the left). The negative charge at  $-x$  creates an electric field at  $P$  pointing toward itself (down and to the left). The vertical ( $y$ ) components of these two fields are equal in magnitude but opposite in direction, so they cancel out. The horizontal ( $x$ ) components both point in the negative  $x$ -direction ( $-\mathbf{i}$ ), so they add together. Hence the net electric field  $\mathbf{E}_p$  points in the  $-\mathbf{i}$  direction.
- (b) The electric potential at  $P$  due to a line element  $dx$  at  $x$  is

$$dV = \frac{1}{4\pi\epsilon_0} \frac{\lambda(x)dx}{\sqrt{x^2 + Y^2}}$$

Integrating over the rod:

$$V_p = \frac{\lambda_0}{4\pi\epsilon_0 a} \int_{-a}^a \frac{x dx}{\sqrt{x^2 + Y^2}}$$

This is an odd function of  $x$ , so the integral vanishes:

$$V_p = 0$$

$\mathbf{E}_p$  cannot be found using  $V_p$  by  $E = -\nabla V_p$  as  $V_p = 0$ .



(c) The dipole moment is

$$\mathbf{p} = \int \mathbf{r} dq = \int_{-a}^a x \mathbf{i} \lambda(x) dx = \mathbf{i} \int_{-a}^a x \frac{\lambda_0 x}{a} dx = \frac{\lambda_0}{a} \mathbf{i} \int_{-a}^a x^2 dx$$

Compute the integral:

$$\int_{-a}^a x^2 dx = \frac{2a^3}{3}$$

Hence, the dipole moment:

$$\mathbf{p} = \frac{2\lambda_0 a^2}{3} \mathbf{i}$$

(d) The potential energy of a dipole in an external field is

$$U = -\mathbf{p} \cdot \mathbf{E}$$

Here,  $\mathbf{E} = E_0(-\mathbf{i} + \mathbf{k})$  and  $\mathbf{p} = \frac{2\lambda_0 a^2}{3} \mathbf{i}$ . Initially,

$$U_i = -\mathbf{p} \cdot \mathbf{E} = -\frac{2\lambda_0 a^2}{3}(-E_0) = \frac{2\lambda_0 a^2 E_0}{3}$$

It will rotate until it reaches a stable equilibrium. This occurs when the dipole moment  $\mathbf{p}$  is aligned with the external field. At this moment,  $U_f = -p|\mathbf{E}| = -\frac{2\sqrt{2}}{3}\lambda_0 a^2 E_0$ . Work done by the field is the negative change in potential energy:

$$W = \frac{2}{3}\lambda_0 a^2 E_0(1 + \sqrt{2})$$

## 3 Magnetic Dipole and Magnetization

### 3.1 Magnetic Dipole

A **magnetic dipole** is a current loop or a pair of magnetic poles that produces a magnetic field similar to that of a small bar magnet. The dipole is characterized by its **magnetic dipole moment**. For a current loop carrying current  $I$  and having area vector  $\mathbf{A}$ , the magnetic dipole moment is defined as

$$\mathbf{m} = I\mathbf{A}$$



where  $\mathbf{A}$  is a vector perpendicular to the plane of the loop with magnitude equal to the area of the loop and direction given by the right-hand rule.

A magnetic dipole placed in a uniform magnetic field  $\mathbf{B}$  experiences a torque given by

$$\boldsymbol{\tau} = \mathbf{m} \times \mathbf{B}$$

This torque tends to align the dipole moment with the magnetic field. The potential energy of a magnetic dipole in a magnetic field is

$$U = -\mathbf{m} \cdot \mathbf{B}$$

This indicates that the energy is minimum when the dipole aligns with the field.

## 3.2 Magnetization

$\mathbf{M}$  is a measure of the magnetic dipole moment per unit volume in a material:

$$\mathbf{M} = \frac{\sum_i \mathbf{m}_i}{V}$$

where  $\mathbf{m}_i$  are the magnetic dipole moments of the atoms or molecules in a volume  $V$ . The magnetization contributes to the total magnetic field in a material. The magnetic field  $\mathbf{B}$  inside a material is related to the applied magnetic field  $\mathbf{H}$  and magnetization  $\mathbf{M}$  by:

$$\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M})$$

The response of a material to an applied magnetic field is often linear for small fields:

$$\mathbf{M} = \chi_m \mathbf{H}$$

where  $\chi_m$  is the **magnetic susceptibility** of the material.



## 4 Maxwell's Equation

### 4.1 Differential Form

**Review of Divergence and Its Physical Meaning** For a vector field  $\mathbf{F}(\mathbf{r})$ , the divergence is defined as

$$\nabla \cdot \mathbf{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}$$

Divergence measures the **net outflow of a vector field** from an infinitesimal point. If  $\nabla \cdot \mathbf{F} > 0$ , the point acts as a **source** of the field. If  $\nabla \cdot \mathbf{F} < 0$ , the point acts as a **sink** of the field. If  $\nabla \cdot \mathbf{F} = 0$ , the field has no net flux coming out.

**Review of Curl and Its Physical Meaning** The **curl** of  $\mathbf{F}$  is defined as

$$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{x} & \mathbf{y} & \mathbf{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix} = \left( \frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z}, \frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x}, \frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \right)$$

The most important physical meaning of curl is that it measures the **local circulation per unit area** of a vector field. Consider a small surface element  $\Delta S$  with unit normal  $\mathbf{n}$ . The circulation of  $\mathbf{F}$  around its boundary  $\partial S$  is

$$\oint_{\partial S} \mathbf{F} \cdot d\mathbf{l}$$

The curl satisfies

$$(\nabla \times \mathbf{F}) \cdot \mathbf{n} = \lim_{\Delta S \rightarrow 0} \frac{1}{\Delta S} \oint_{\partial S} \mathbf{F} \cdot d\mathbf{l}.$$

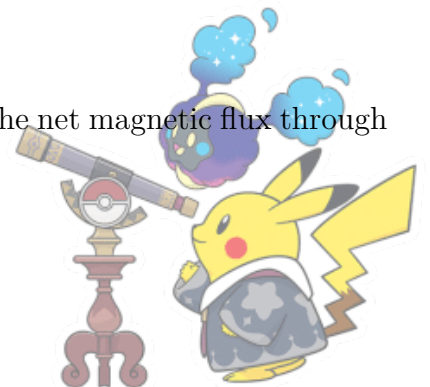
**Gauss's Law of Electrostatics** Gauss's law relates the electric field  $\mathbf{E}$  to the electric charge density  $\rho$ :

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}$$

Integral form follows by the Gauss's theorem in Multi-variable Calculus.

**Gauss's Law for Magnetism** There are no magnetic monopoles. Hence, the net magnetic flux through any closed surface is zero:

$$\nabla \cdot \mathbf{B} = 0$$



Integral form follows by the Gauss's theorem in Multi-variable Calculus.

### Ampère-Maxwell Law

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t}$$

The second term is the **displacement current** which we will discuss later. Magnetic fields circulate around

- electric currents  $\mathbf{J}$ , and
- time-varying electric fields.

Both act as sources of magnetic curl.

### Faraday's Law of Induction

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

A changing magnetic field produces a **nonzero curl of the electric field**. This means the electric field forms **closed loops** rather than pointing from charges.

## 4.2 Integral Form

**Gauss's Law of Electrostatics** Gauss's Law relates the electric flux through a closed surface to the charge enclosed by that surface. It is one of Maxwell's equations and is especially useful for calculating electric fields of highly symmetric charge distributions. Mathematically, Gauss's Law states

$$\oint_{\mathcal{S}} \mathbf{E} \cdot d\mathbf{A} = \frac{Q_{\text{enc}}}{\varepsilon_0}$$

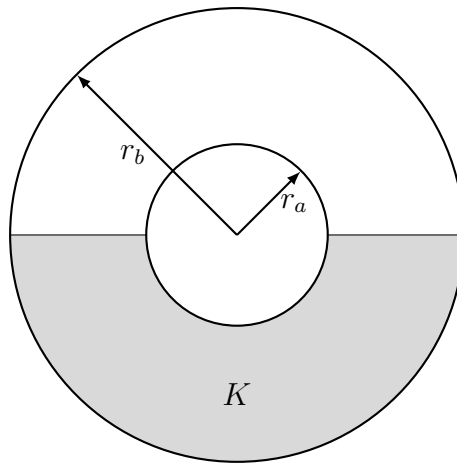
where

- $\mathbf{E}$  is the electric field vector,
- $d\mathbf{A}$  is a vector representing an infinitesimal area on the closed surface  $\mathcal{S}$ , pointing outward,
- $Q_{\text{enc}}$  is the total charge enclosed within the surface  $\mathcal{S}$ , and
- $\varepsilon_0$  is the vacuum permittivity ( $\varepsilon_0 \approx 8.854 \times 10^{-12}$  F/m).

**Example.**







Consider a spherical capacitor which has total charge  $+Q$  on its inner conductor (radius  $r_a$ ) and charge  $-Q$  on the outer conductor (radius  $r_b$ ).

- (a) Using Gauss's law, determine the electric field  $E_0(r)$  as a function of  $r$ , the distance from the center of the capacitor. Hence determine the capacitance  $C_0$  of this capacitor.
- (b) Now half of the interior of the capacitor is filled with a liquid of dielectric constant  $K$ , shown below in Figure 3. Determine the capacitance of the half-filled capacitor. (Hint: Treat this as a combination of two capacitors of hemispheres)
- (c) Determine the electric field  $E(r)$ , together with the free ( $\sigma_f$ ) and bound ( $\sigma_b$ ) surface charge densities on the inner sphere, for both the upper and lower portion of the capacitor.
- (a) Note that when  $r < r_a$  and  $r > r_b$ ,  $E_0(r) = 0$ . Applying Gauss's Law  $\oint \mathbf{E} \cdot d\mathbf{A} = \frac{Q_{enc}}{\epsilon_0}$  for a Gaussian sphere of radius  $r$  ( $r_a < r < r_b$ ):

$$E_0(r) \cdot 4\pi r^2 = \frac{Q}{\epsilon_0} \implies E(r) = \frac{Q}{4\pi\epsilon_0 r^2}$$

The potential difference  $V$  between the conductors is

$$V = - \int_{r_b}^{r_a} E_0 dr = \int_{r_a}^{r_b} \frac{Q}{4\pi\epsilon_0 r^2} dr = \frac{Q}{4\pi\epsilon_0} \left( \frac{1}{r_a} - \frac{1}{r_b} \right) = \frac{Q(r_b - r_a)}{4\pi\epsilon_0 r_a r_b}$$

The capacitance  $C_0$  is defined as  $Q/V$ :

$$C_0 = \frac{4\pi\epsilon_0 r_a r_b}{r_b - r_a}$$

- (b) Note that



- Vacuum hemisphere:  $C_1 = \frac{C_0}{2}$
- Dielectric hemisphere:  $C_2 = \frac{KC_0}{2}$

Hence,  $C = 2\pi\epsilon_0(1 + K)\frac{r_a r_b}{r_b - r_a}$ .

(c) The potential difference  $V$  is the same for both regions. Since  $E = -\frac{dV}{dr}$ , the radial electric field must be the same in both the vacuum and dielectric regions:

$$V = \frac{Q}{C} = \frac{Q(r_b - r_a)}{2\pi\epsilon_0 r_a r_b (1 + K)}$$

The electric field  $E(r)$  for  $r_a < r < r_b$  is

$$E(r) = \frac{Q}{2\pi\epsilon_0(1 + K)r^2}$$

The free charge density is given by  $\sigma_f = \mathbf{D} \cdot \mathbf{n}$ . At the inner conductor ( $r = r_a$ ):

- Upper Portion (Vacuum):  $D = \epsilon_0 E(r_a)$

$$\sigma_{f,\text{upper}} = \frac{Q}{2\pi(1 + K)r_a^2}$$

- Lower Portion (Dielectric):  $D = \epsilon_0 K E(r_a)$

$$\sigma_{f,\text{lower}} = \frac{KQ}{2\pi(1 + K)r_a^2}$$

Bound charge exists only where there is a dielectric. It is defined as  $\sigma_b = \mathbf{P} \cdot \mathbf{n}_{\text{out}}$ , where  $\mathbf{n}_{\text{out}}$  is the unit normal pointing out of the dielectric. At the inner surface, this normal points toward the center ( $-\mathbf{r}$ ).

- Upper Portion: No dielectric, so  $\sigma_{b,\text{upper}} = 0$ .
- Lower Portion: Using  $\mathbf{P} = \epsilon_0(K - 1)\mathbf{E}$ , we have

$$\sigma_{b,\text{lower}} = \mathbf{P} \cdot (-\mathbf{r}) = -\epsilon_0(K - 1)E(r_a)$$

$$\sigma_{b,\text{lower}} = -\frac{(K - 1)Q}{2\pi(1 + K)r_a^2}$$



**Example.** An insulating cylinder of radius  $a$  and length  $L$  has a uniform volume charge density  $\rho$  and a total charge  $+2Q$ . Concentric with the inner cylinder is a conducting hollow cylindrical shell of the same length with total charge  $-Q$ , whose inner and outer radii are  $b$  and  $c$  respectively. Assume that

$$a, b, c \ll L$$

so that end effects can be neglected.

- Explain, using symmetry arguments, why the electric field of an infinitely long cylindrical conductor cannot have a  $\theta$ -component in cylindrical coordinates (i.e.  $E_\theta = 0$ ).
- Determine the surface charge densities on the inner and outer surfaces of the conducting shell at radii  $r = b$  and  $r = c$ .
- Using Gauss's law, find the magnitude of the electric field in the regions

$$r < a, \quad a < r < b, \quad b < r < c, \quad r > c.$$

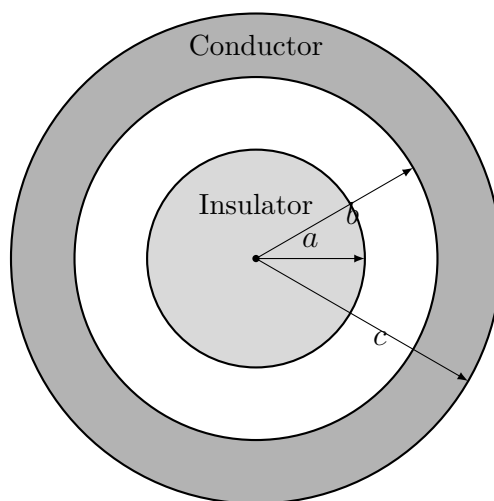


Figure 1: Cross-sectional view of the charged insulating cylinder and concentric conducting shell.

- Assume to the contrary that there is a  $\theta$ -component.
  - If  $\theta$ -component is uniform, then for reflection  $\theta \rightarrow -\theta$ ,  $E_\theta$  would change sign, which violates reflectional symmetry.
  - If it is not uniform, then it can select a preferred angular direction, which violates rotational symmetry.



(b) Note that the charge induced on the inner surface must cancel the net charge enclosed inside the conductor. Therefore,  $Q_{\text{inner}} = -2Q$ . Then  $Q_{\text{outer}} = +Q$ . Therefore,

$$\sigma_b = \frac{-2Q}{2\pi bL}, \quad \sigma_c = \frac{+Q}{2\pi bL}$$

(c) Use a cylindrical Gaussian surface of radius  $r$  and length  $L$ . Consider the following cases:

- **Case 1:**  $r < a$

$$E(2\pi rL) = \frac{\rho\pi r^2L}{\varepsilon_0} \implies E(r) = \frac{\rho r}{2\varepsilon_0}$$

Note that  $\rho = \frac{2Q}{\pi a^2L}$ . Therefore,  $E(r) = \frac{Qr}{\pi\varepsilon_0 a^2L}$ .

- **Case 2:**  $a < r < b$

$$E(2\pi rL) = \frac{2Q}{\varepsilon_0} \implies E(r) = \frac{Q}{\pi\varepsilon_0 Lr}$$

- **Case 3:**  $b < r < c$

Inside the conductor,  $E = 0$ .

- **Case 4:**  $r > c$

Total enclosed charge =  $2Q + (-Q) = Q$ .

$$E(2\pi rL) = \frac{Q}{\varepsilon_0}$$

Therefore,  $E(r) = \frac{Q}{2\pi\varepsilon_0 Lr}$ .

**Gauss's Law for Magnetism** Let  $\mathbf{B}$  be the magnetic field vector, and let  $S$  be a closed surface. Gauss's Law for Magnetism is expressed as

$$\oint_S \mathbf{B} \cdot d\mathbf{A} = 0$$

where  $d\mathbf{A}$  is the vector area element of the surface  $S$ .

**Ampère-Maxwell Law** In classical electromagnetism, Ampère's law relates the magnetic field  $\mathbf{B}$  around a closed loop to the electric current  $\mathbf{J}$  passing through the loop:

$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{enc}}$$



where  $I_{\text{enc}}$  is the current enclosed by the loop, and  $\mu_0$  is the permeability of free space. However, this form of Ampère's law fails when dealing with time-varying electric fields, such as between the plates of a charging capacitor. To fix this, Maxwell introduced the concept of displacement current. Note that electric field between capacitor's plates is given by  $E = \frac{\sigma}{\epsilon_0} = \frac{Q}{\epsilon A}$ . Then  $Q = \epsilon_0 \Phi_E$ . We can define  $\frac{dQ}{dt}$  to be displacement current and the modified Ampère-Maxwell law becomes

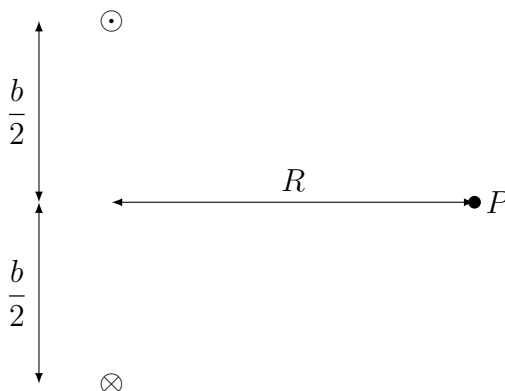
$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 \left( I_{\text{enc}} + \epsilon_0 \frac{d\Phi_E}{dt} \right)$$

where  $\Phi_E$  is the electric flux through the surface bounded by the loop:

$$\Phi_E = \int \mathbf{E} \cdot d\mathbf{A}$$

**Example.** Consider two very long straight wires a distance  $b$  apart carrying equal and opposite currents  $I$  as shown above. The upper wire carries current **out of the page** ( $\odot$ ), while the lower wire carries current **into the page** ( $\otimes$ ).

- Explain why the magnetic field due to an infinitely long current-carrying wire **cannot** have a radial component in cylindrical coordinates. Assume the current flows along the  $z$ -direction. (Hint: Use either the Biot-Savart law or symmetry / Gauss' law for magnetism.)
- Using Ampère-Maxwell Law, determine the magnetic field (magnitude and direction) at a distance  $d$  from a very long wire carrying current  $I$ .
- Calculate the magnetic field (magnitude and direction) at point  $P$ , which is equidistant from the two wires. For large  $R$ , the magnetic field can be shown to be proportional to  $R^\alpha$ . Determine  $\alpha$ .



- There are two methods:



- The Biot–Savart law gives

$$d\mathbf{B} = \frac{\mu_0}{4\pi} \frac{I d\boldsymbol{\ell} \times \mathbf{r}}{r^2}$$

Here  $d\boldsymbol{\ell} \parallel \mathbf{z}$ , so  $d\boldsymbol{\ell} \times \mathbf{r}$  is purely azimuthal ( $\phi$ ). Thus the magnetic field has **only** a  $\phi$ -component:

$$\mathbf{B} = B_\phi(r) \boldsymbol{\phi}$$

- Consider a cylindrical Gaussian surface of radius  $r$  and length  $L$  centered on the wire. The area vector  $d\mathbf{A}$  points radially outward ( $\mathbf{r}$ ). If there is a radial field component  $B_r$ , the flux is  $\int B_r \cdot d\mathbf{A} = B_r(2\pi rL)$ . As the wire is infinite, the magnetic field is translationally invariant along  $z$ . This means the flux entering one end cap is exactly equal to the flux leaving the other. Their contributions to the total flux cancel out. Applying Gauss's Law:

$$B_r(2\pi rL) + 0 = 0$$

The only solution is:

$$B_r = 0$$

- (b) Choose a circular Amperian loop of radius  $d$  centered on the wire. By symmetry,  $\mathbf{B}$  is tangential and constant on the loop. Ampère-Maxwell Law gives

$$\oint \mathbf{B} \cdot d\boldsymbol{\ell} = B(2\pi d) = \mu_0 I$$

Therefore,

$$B(d) = \frac{\mu_0 I}{2\pi d}$$

with direction given by the right-hand rule (azimuthal around the wire).

- (c) Each wire is a distance

$$r = \sqrt{R^2 + \left(\frac{b}{2}\right)^2}$$

from point  $P$ . The magnitude of the field due to each wire is

$$B_0 = \frac{\mu_0 I}{2\pi r}$$

At point  $P$ , the horizontal components of the two magnetic fields cancel by symmetry, while the





vertical components add. Let  $\theta$  be the angle between  $\mathbf{r}$  and the horizontal:

$$\sin \theta = \frac{b/2}{r}$$

The vertical component from one wire is

$$B_y^{(1)} = B_0 \sin \theta = \frac{\mu_0 I}{2\pi r} \cdot \frac{b/2}{r}$$

Thus the total field is

$$B_{\text{net}} = 2B_y^{(1)} = \frac{\mu_0 I b}{2\pi r^2}$$

For  $R \gg b$ ,

$$r^2 \approx R^2$$

so

$$B_{\text{net}} \sim \frac{\mu_0 I b}{2\pi R^2}$$

and  $\alpha = -2$ .

**Faraday's Law of Electromagnetic Induction** Faraday's law states that the induced electromotive force (emf) around a closed loop equals the negative rate of change of magnetic flux through the surface bounded by the loop:

$$\oint_{\partial S} \mathbf{E} \cdot d\mathbf{l} = -\frac{d}{dt} \int_S \mathbf{B} \cdot d\mathbf{A}$$

where

- $\partial S$  is the closed conducting loop,
- $S$  is any surface bounded by the loop,
- $\mathbf{E}$  is the electric field, and
- $\mathbf{B}$  is the magnetic field.

The magnetic flux through a surface  $S$  is

$$\Phi_B = \int_S \mathbf{B} \cdot d\mathbf{A}$$



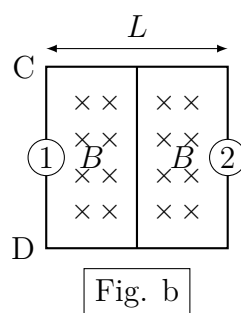
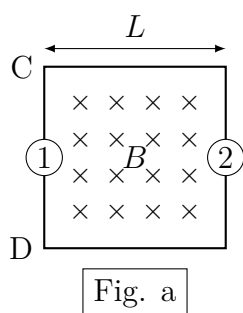
Hence Faraday's law may be written compactly as

$$\mathcal{E} = -\frac{d\Phi_B}{dt}$$

as  $\mathbf{E} \cdot d\mathbf{l}$  is the work done per unit charge by the electric field when a test charge moves an infinitesimal distance  $d\mathbf{l}$  which is the definition of the emf. The negative sign expresses Lenz's law:

*The induced current produces a magnetic field that opposes the change in magnetic flux.*

**Example.** A square conducting loop of side  $L$  contains two identical lightbulbs, 1 and 2, as shown in Fig. a. A magnetic field exists in the region inside the loop with magnitude as a function of time  $t$  given by  $B(t) = at + b$ , where  $a$  and  $b$  are positive constants. The lightbulbs each have a resistance  $R_0$ .



- Determine the induced emf in the loop.
- Determine the induced current in the loop. Compare the electric potentials at the corners C and D. Explain your answer.
- Using results from parts (a) and (b), determine the power dissipated in light bulb 1.
- Suppose a wire is now added to connect the midpoints of the top and bottom of the loop, as shown in Fig b. Would the brightness of bulb 1 be brighter, dimmer, or identical to that before the connection? Explain your answer.
- The magnetic flux through the loop is given by  $\Phi_B = \mathbf{B} \cdot \mathbf{A} = B(t)L^2$ . Using Faraday's Law,

$$\mathcal{E} = \left| \frac{d\Phi_B}{dt} \right| = \frac{d}{dt} ((at + b)L^2) = aL^2$$

- The total resistance of the loop is the sum of the two lightbulbs in series:  $R_{total} = 2R_0$ . The induced current is

$$I = \frac{\mathcal{E}}{R_{total}} = \frac{aL^2}{2R_0}$$



By Lenz's Law, as the magnetic field into the page is increasing ( $a > 0$ ), the induced current flows **counter-clockwise** to create a field pointing out of the page.

Regarding the potentials, note that this is not an electrostatic situation as the electric field is non-conservative due to the changing magnetic field by

$$\oint \mathbf{E} \cdot d\mathbf{l} \neq 0$$

As the loop is perfectly symmetric and the two bulbs are identical, the induced emf is distributed evenly. The work done by the induced electric field in moving a charge from point D to point C (half the loop) is

$$\frac{1}{2}\mathcal{E}$$

The voltage drop across bulb 1 is

$$IR_0 = \left(\frac{\mathcal{E}}{2R_0}\right)R_0 = \frac{1}{2}\mathcal{E}$$

Using the line integral of the total electric field  $\int_D^C \mathbf{E}_{total} \cdot d\mathbf{l} = \int_D^C \mathbf{E}_{induced} \cdot d\mathbf{l} - (V_C - V_D) = IR_0$ ,  $(V_C - V_D) = 0$ .

(c) The power dissipated in bulb 1 is given by  $P = I^2 R_0$ :

$$P_1 = \left(\frac{aL^2}{2R_0}\right)^2 R_0 = \frac{a^2 L^4}{4R_0}$$

(d) Note that

- The area of the left loop is now  $A_{left} = \frac{L^2}{2}$ .
- The induced emf in this specific loop is  $\mathcal{E}_{left} = \frac{d}{dt}(B \cdot \frac{L^2}{2}) = \frac{aL^2}{2}$ .
- Consequently, bulb 1 is the only resistance in the left loop. The current through it is

$$I_{new} = \frac{\mathcal{E}_{left}}{R_0} = \frac{aL^2/2}{R_0} = \frac{aL^2}{2R_0}$$

As the current is the same as in part (b), the brightness of bulb 1 will be **identical**.



## 5 Circuit

### 5.1 Kirchhoff's Circuit Law

**Kirchhoff's Current Law (KCL)** states that

*At any node (junction) in an electrical circuit, the algebraic sum of currents is zero.*

Mathematically,

$$\sum_k I_k = 0$$

where currents entering the node are taken as positive and currents leaving the node as negative (or vice versa, consistently).

**Proof.** Consider a small volume enclosing a circuit node. Let  $Q$  be the total charge inside. The rate of change of charge is

$$\frac{dQ}{dt}$$

The net current flowing out of the volume is

$$\sum I_{\text{out}} - \sum I_{\text{in}}$$

Charge conservation requires

$$\frac{dQ}{dt} = \sum I_{\text{in}} - \sum I_{\text{out}}$$

For lumped-element circuits under steady conditions,

$$\frac{dQ}{dt} = 0$$

so

$$\sum I_{\text{in}} = \sum I_{\text{out}}.$$

**Kirchhoff's Voltage Law** states that

*The algebraic sum of all voltages around any closed loop in a circuit is equal to zero.*

Mathematically, for a loop with  $m$  elements:

$$\sum_{k=1}^m V_k = 0$$

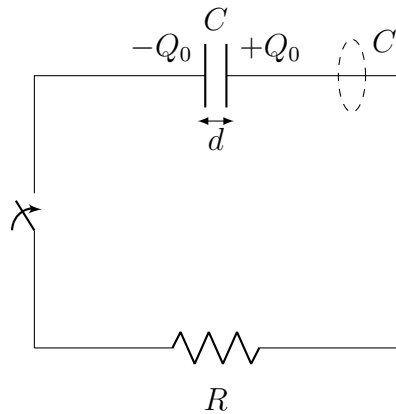


Where  $V_k$  is the voltage across element  $k$ . Voltage rises are taken as positive and drops as negative (or vice versa).

Consider a simple loop with a battery of voltage  $V$  and two resistors  $R_1$  and  $R_2$  in series. If  $I$  is the current in the loop, Kirchhoff's Voltage Law gives

$$V - IR_1 - IR_2 = 0 \implies I = \frac{V}{R_1 + R_2}$$

**Example.** A capacitor  $C$  is made of two circular plates of radius  $r$  separated by a distance  $d$ , and it is connected to a resistor  $R$  as shown. The capacitor is initially charged with  $Q_0$  on its plates as shown below. Define clockwise currents as positive.



- At time  $t = 0$ , the switch is closed. Using Kirchhoff's rules, set up the differential equation for  $Q(t)$ , the charge on the positive capacitor plate. Show that the solution is in the form  $Q(t) = Q_0 e^{-t/X}$  and determine  $X$ . Hence also determine the current  $I(t)$  in the circuit.
  - Using the loop  $C'$  surrounding the current wire as shown in the figure above as an example, does the current  $I(t)$  you determined in part (a) satisfy Ampere's Law? Explain your answer.
  - By finding the displacement current inside the capacitor plates, determine the induced magnetic field in the space between the capacitor plates as a function of  $x$ , the distance from the center of the capacitor plates, for both  $x < r$  and  $x > r$ .
- (a) According to Kirchhoff's Loop Rule, the sum of the potential differences around the closed loop must be zero:

$$V_C - V_R = 0 \implies \frac{Q(t)}{C} - I(t)R = 0$$



Since the clockwise current  $I(t)$  flows away from the positive plate, the current is equal to the rate of decrease of the charge on that plate:

$$I(t) = -\frac{dQ}{dt}$$

Substituting this into the loop equation:

$$\frac{Q}{C} - \left(-\frac{dQ}{dt}\right)R = 0 \implies R\frac{dQ}{dt} + \frac{1}{C}Q = 0$$

$$\frac{dQ}{dt} = -\frac{1}{RC}Q$$

Integrating from  $t = 0$  (where  $Q = Q_0$ ) to  $t$ :

$$\int_{Q_0}^Q \frac{dQ'}{Q'} = -\frac{1}{RC} \int_0^t dt' \implies \ln\left(\frac{Q(t)}{Q_0}\right) = -\frac{t}{RC}$$

$$Q(t) = Q_0 e^{-t/RC}$$

Comparing this to  $Q(t) = Q_0 e^{-t/X}$ , we find:

$$X = RC$$

The current  $I(t)$  is

$$I(t) = -\frac{dQ}{dt} = -\left(-\frac{Q_0}{RC}e^{-t/RC}\right) = \frac{Q_0}{RC}e^{-t/RC}$$

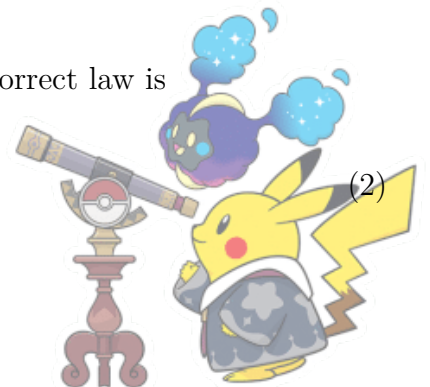
- (b) Consider the loop  $C'$  that surrounds the wire but passes between the capacitor plates. For this surface, no conduction current passes through it. Using the original Ampère-Maxwell Law,

$$\oint \mathbf{B} \cdot d\boldsymbol{\ell} = \mu_0 I_{\text{enc}} \quad (1)$$

one would obtain  $I_{\text{enc}} = 0$ , implying a vanishing magnetic field, which contradicts the magnetic field produced by the current in the circuit.

This apparent inconsistency is resolved by Maxwell's correction. The correct law is

$$\oint \mathbf{B} \cdot d\boldsymbol{\ell} = \mu_0 \left( I_{\text{cond}} + \epsilon_0 \frac{d\Phi_E}{dt} \right) \quad (2)$$



Between the capacitor plates, the displacement current

$$I_d = \epsilon_0 \frac{d\Phi_E}{dt}$$

replaces the conduction current, ensuring consistency of Ampère-Maxwell Law.

- (c) Inside the capacitor, there is no conduction current, but there is a changing electric field  $E$ . The electric field between the plates is:

$$E = \frac{\sigma}{\epsilon_0} = \frac{Q(t)}{\epsilon_0 \pi r^2}$$

The displacement current  $I_d$  is defined by the change in electric flux  $\Phi_E$ :

$$I_d = \epsilon_0 \frac{d\Phi_E}{dt}$$

Case 1:  $x < r$  (Inside the plates)

The flux through a circle of radius  $x$  is  $\Phi_E = E \cdot \pi x^2 = \frac{Q(t)}{\epsilon_0 r^2} x^2$ .

$$I_{d,enclosed} = \epsilon_0 \frac{d}{dt} \left( \frac{Q(t)x^2}{\epsilon_0 r^2} \right) = \frac{x^2}{r^2} \frac{dQ}{dt} = -\frac{x^2}{r^2} I(t)$$

Using the Ampère-Maxwell Law  $\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{d,enclosed}$ :

$$B(2\pi x) = \mu_0 \left( \frac{x^2}{r^2} I(t) \right)$$

$$B(x) = \frac{\mu_0 I(t) x}{2\pi r^2} \quad \text{for } x < r$$

Case 2:  $x > r$  (Outside the plates)

The total displacement current is enclosed, which equals the conduction current in magnitude:

$$I_{d,total} = \frac{dQ}{dt} = -I(t).$$

$$B(2\pi x) = \mu_0 I(t)$$

$$B(x) = \frac{\mu_0 I(t)}{2\pi x} \quad \text{for } x > r$$





## 5.2 Capacitance

**Capacitance** is a measure of the ability of a system to store electric charge per unit potential difference. It is defined as

$$C = \frac{Q}{\Delta V}$$

where

- $Q$  is the magnitude of charge stored on one conductor,
- $\Delta V$  is the potential difference between conductors.

**Example.** To find the capacitance of an isolated sphere, consider a system of two concentric spherical shells: an inner shell of radius  $r$  and an outer shell of radius  $R_{outer}$ .

- The potential difference  $\Delta V$  between the shells is given by:

$$\Delta V = \frac{Q}{4\pi\epsilon_0} \left( \frac{1}{r} - \frac{1}{R_{outer}} \right)$$

- Using the definition of capacitance  $C = \frac{Q}{\Delta V}$ :

$$C = \frac{Q}{\frac{Q}{4\pi\epsilon_0} \left( \frac{1}{r} - \frac{1}{R_{outer}} \right)} = \frac{4\pi\epsilon_0}{\frac{1}{r} - \frac{1}{R_{outer}}}$$

- To find the capacitance of an isolated sphere, we allow the radius of the outer shell to go to infinity ( $R_{outer} \rightarrow \infty$ ):

$$C = \lim_{R_{outer} \rightarrow \infty} \frac{4\pi\epsilon_0}{\frac{1}{r} - \frac{1}{R_{outer}}} = 4\pi\epsilon_0 r$$

The SI unit of capacitance is the **farad (F)**, where

$$1 \text{ F} = 1 \text{ C V}^{-1}$$

Consider two parallel conducting plates of area  $A$ , separated by distance  $d$ , filled with a dielectric of permittivity  $\epsilon$ . The electric field between the plates is

$$E = \frac{V}{d}$$



Using Gauss's law,

$$E = \frac{\sigma}{\epsilon}$$

where  $\sigma = Q/A$  is the surface charge density. Combining the equations,

$$V = \frac{Qd}{\epsilon A}$$

Hence, the capacitance is

$$C = \frac{\epsilon A}{d}$$

If a dielectric with relative permittivity  $\kappa$  is inserted,

$$\epsilon = \kappa \epsilon_0$$

so the capacitance increases to

$$C = \kappa C_0$$

The work done to charge a capacitor is stored as electric potential energy:

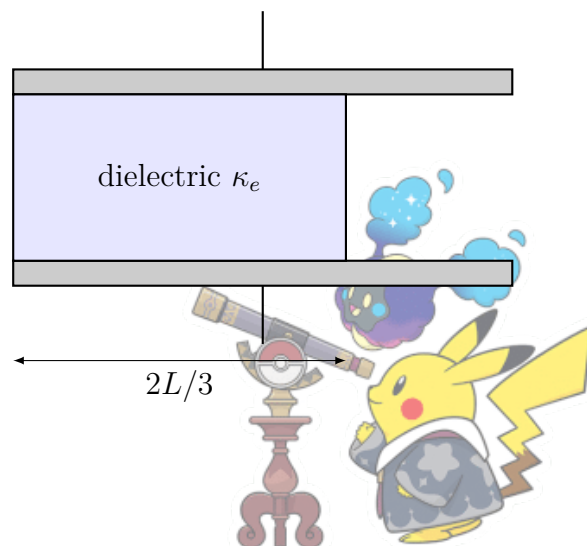
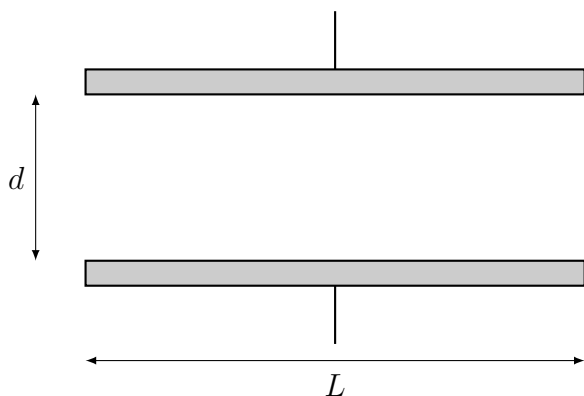
$$U = \frac{1}{2}CV^2 = \frac{Q^2}{2C}.$$

In terms of the electric field,

$$u_E = \frac{1}{2}\epsilon E^2$$

is the energy density of the electric field.

**Example.** Consider a parallel-plate capacitor whose plates are metal squares of side length  $L$ , separated by a distance  $d$ . The capacitor is connected to a battery of constant voltage  $\mathcal{E}$  for a long time, so that a total free charge  $+Q$  resides on the top plate and  $-Q$  on the bottom plate.



- (a) A student (with appropriate protection) pushes a dielectric slab of dielectric constant  $\kappa_e$  into the capacitor from the right, as shown above. Explain whether
- (i) the free charge on the capacitor plates, and
  - (ii) the energy stored in the capacitor
- increase, decrease, or remain constant as the dielectric is inserted.
- (b) Suppose the dielectric is pushed **two-thirds of the way** into the capacitor. Calculate the **total work** done by the student. Indicate whether the work is positive or negative.
- (c) Sketch the distribution and relative amount of **free** and **induced** charges on the capacitor in the right-hand figure, assuming  $\kappa_e = 2$ . A higher charge density should be indicated by a higher density of the symbols  $+$  and  $-$ .
- (a) (i) By  $Q = C\mathcal{E}$ , as the dielectric is inserted, the capacitance increases, so  $Q$  increases.
- (ii) For a capacitor held at constant voltage,

$$U = \frac{1}{2}C\mathcal{E}^2$$

As the dielectric is inserted, the capacitance increases, so the energy stored in the capacitor also increases.

- (b) The dielectric fills a fraction of the plate area. Since the dielectric is inserted laterally, the capacitor behaves as two capacitors in **parallel**:

- Dielectric-filled region: area  $A_d = \frac{2}{3}L^2$
- Vacuum region: area  $A_v = \frac{1}{3}L^2$

Their capacitances are

$$C_d = \kappa_e \epsilon_0 \frac{A_d}{d} = \kappa_e \epsilon_0 \frac{2L^2}{3d}$$

$$C_v = \epsilon_0 \frac{A_v}{d} = \epsilon_0 \frac{L^2}{3d}$$

Hence, the total capacitance is

$$C_f = \epsilon_0 \frac{L^2}{d} \left( \frac{1}{3} + \frac{2\kappa_e}{3} \right)$$



The initial capacitance (vacuum-filled) is

$$C_i = \varepsilon_0 \frac{L^2}{d}$$

As the voltage is fixed,

$$\begin{aligned} \Delta U &= \frac{1}{2} \mathcal{E}^2 (C_f - C_i) \\ &= \frac{1}{2} \varepsilon_0 \frac{L^2}{d} \mathcal{E}^2 \left( \frac{2\kappa_e - 2}{3} \right) \end{aligned}$$

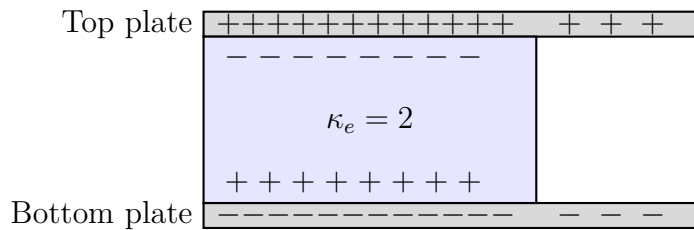
The dielectric is **pulled into** the capacitor by the electric field. Therefore, the student must exert a force opposite to the motion. The work done by the student equals the negative of the work done by the electric field:

$$W_{\text{student}} = -\Delta U$$

Hence,

$$W_{\text{student}} = -\frac{1}{2} \varepsilon_0 \frac{L^2}{d} \mathcal{E}^2 \left( \frac{2\kappa_e - 2}{3} \right)$$

(c)



Note that for free charge density,

- $\sigma_{\text{air}} = \varepsilon_0 \frac{\mathcal{E}}{d}$
- $\sigma_{\text{dielectric}} = \kappa_e \varepsilon_0 \frac{\mathcal{E}}{d}$

which differ by the factor of  $\kappa_e$ .

The displacement field  $\mathbf{D}$  in the dielectric is defined as

$$\mathbf{D} = \varepsilon_0 \mathbf{E}_{\text{net}} + \mathbf{P}$$



where  $\mathbf{P}$  is the polarization of the dielectric. By Gauss's Law,

$$\nabla \cdot \mathbf{E}_0 = \frac{\rho_{\text{free}}}{\varepsilon_0}, \quad \nabla \cdot \mathbf{D} = \rho_{\text{free}}$$

Then  $\mathbf{D} = \varepsilon_0 \mathbf{E}_0$ .

For a linear dielectric,

$$\mathbf{P} = \varepsilon_0(\kappa_e - 1)\mathbf{E}_{\text{net}}$$

Substitute  $\mathbf{P}$  into  $\mathbf{D}$ :

$$\mathbf{D} = \varepsilon_0 \mathbf{E}_{\text{net}} + \varepsilon_0(\kappa_e - 1)\mathbf{E}_{\text{net}} = \varepsilon_0 \kappa_e \mathbf{E}_{\text{net}}$$

Then

$$\varepsilon_0 \mathbf{E}_0 = \varepsilon_0 \kappa_e \mathbf{E}_{\text{net}} \quad \implies \quad E_{\text{net}} = \frac{E_0}{\kappa_e}$$

Furthermore,

$$\begin{aligned} E_{\text{net}} &= \frac{E_0}{\kappa_e} \\ \frac{\sigma_f - \sigma_b}{\varepsilon_0} &= \frac{\sigma_f}{\kappa_e \varepsilon_0} \\ \sigma_f - \sigma_b &= \frac{\sigma_f}{\kappa_e} \\ \sigma_b &= \sigma_f - \frac{\sigma_f}{\kappa_e} \\ \sigma_b &= \sigma_f \left( 1 - \frac{1}{\kappa_e} \right) \end{aligned}$$

### 5.3 Inductance

**Inductance** measures the ability of a circuit to oppose changes in current due to induced magnetic fields.

It is defined by

$$\Phi_B = LI$$

where

- $\Phi_B$  is the magnetic flux linkage,
- $I$  is the current,
- $L$  is the inductance.



The SI unit of inductance is the **henry (H)**, where

$$1 \text{ H} = 1 \text{ Wb A}^{-1}$$

A changing current produces an induced emf:

$$\mathcal{E} = -\frac{d\Phi_B}{dt}$$

Using  $\Phi_B = LI$ ,

$$\mathcal{E} = -L\frac{dI}{dt}$$

The negative sign represents **Lenz's law**.

The energy stored in the magnetic field is

$$U = \frac{1}{2}LI^2$$

The magnetic energy density is

$$u_B = \frac{B^2}{2\mu}$$

**Example.** For a solenoid of  $N$  turns, length  $\ell$ , cross-sectional area  $A$ , and permeability  $\mu$ , the magnetic field inside is

$$B = \mu\frac{N}{\ell}I$$

Magnetic flux per turn:

$$\Phi = BA = \mu\frac{N}{\ell}IA$$

Total flux linkage:

$$\Phi_B = N\Phi = \mu\frac{N^2A}{\ell}I$$

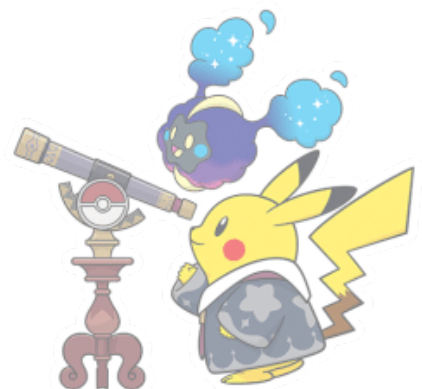
Hence,

$$L = \mu\frac{N^2A}{\ell}$$

## 5.4 AC Circuit

Impedance is a complex quantity defined as

$$Z = R + jX$$



where

- $R$  is the **resistance** (real part), measured in ohms ( $\Omega$ ),
- $X$  is the **reactance** (imaginary part), measured in ohms ( $\Omega$ ),
- $j$  is the imaginary unit ( $j^2 = -1$ ).

Reactance represents the opposition to AC due to energy storage elements (inductors and capacitors).

**Theorem.** The reactance of an inductor  $L$  is

$$X_L = \omega L$$

where  $\omega = 2\pi f$  is the angular frequency of the AC source. The voltage across an inductor leads the current by  $90^\circ$ .

**Theorem.** The reactance of a capacitor  $C$  is

$$X_C = -\frac{1}{\omega C}$$

The voltage across a capacitor lags the current by  $90^\circ$ .

**Properties.**

- Impedance adds like vectors in series:  $Z_{\text{total}} = Z_1 + Z_2 + \dots$
- For parallel components, the total impedance is

$$\frac{1}{Z_{\text{total}}} = \frac{1}{Z_1} + \frac{1}{Z_2} + \dots$$

- Ohm's law in AC circuits becomes

$$\tilde{V}_{\text{in}} = Z_{\text{eq}} \tilde{I}$$

where  $Z_{\text{eq}}$  is the equivalent impedance of the circuit, and  $\tilde{V}_{\text{in}}$  and  $\tilde{I}$  are represented in complex form.

## 5.5 RLC Circuit

### 5.5.1 Equation

An **RLC circuit** consists of a resistor ( $R$ ), inductor ( $L$ ), and capacitor ( $C$ ) connected together. It is the electrical analogue of a **damped harmonic oscillator**. Consider a series circuit driven by a voltage



source  $V(t)$ . Applying Kirchhoff's Voltage Law around the loop:

$$V_R + V_L + V_C = V(t)$$

Using constitutive relations,

$$V_R = IR, \quad V_L = L \frac{dI}{dt}, \quad V_C = \frac{Q}{C}, \quad I = \frac{dQ}{dt}$$

we obtain the governing equation:

$$L \frac{d^2 Q}{dt^2} + R \frac{dQ}{dt} + \frac{1}{C} Q = V(t)$$

For a source-free circuit,

$$L\ddot{Q} + R\dot{Q} + \frac{1}{C}Q = 0$$

Define

$$\omega_0 = \frac{1}{\sqrt{LC}}, \quad \gamma = \frac{R}{2L}$$

The equation becomes

$$\ddot{Q} + 2\gamma\dot{Q} + \omega_0^2 Q = 0$$

- **Underdamped** ( $\gamma < \omega_0$ ):

$$Q(t) = Q_0 e^{-\gamma t} \cos(\omega_d t + \phi), \quad \omega_d = \sqrt{\omega_0^2 - \gamma^2}$$

- **Critically damped** ( $\gamma = \omega_0$ ):

$$Q(t) = (A + Bt)e^{-\omega_0 t}$$

- **Overdamped** ( $\gamma > \omega_0$ ):

$$Q(t) = Ae^{r_1 t} + Be^{r_2 t}$$

### 5.5.2 Boundary Condition

As voltage across a capacitor cannot change instantaneously:

$$Q(0^+) = Q(0^-)$$





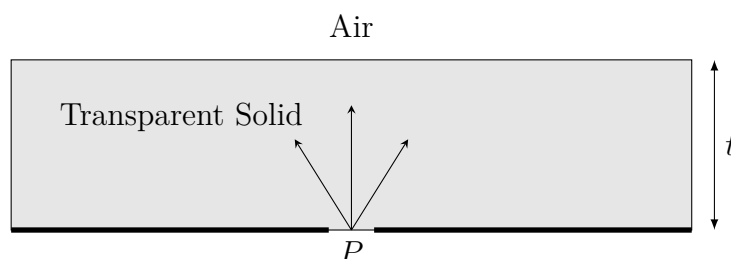
An instantaneous change in current would require infinite voltage, so

$$I(0^+) = I(0^-)$$

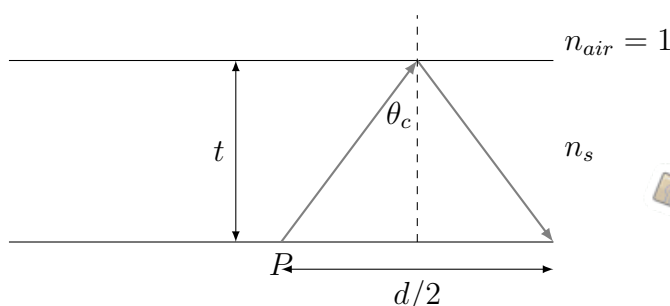
## 6 Optics

### 6.1 Review of DSE Concepts

**Example.** A method for measuring the index of refraction  $n_s$  of a transparent solid is shown below. One face of a slab of thickness  $t$  is painted white, and a small hole scraped clear at point  $P$  serves as a source of diverging rays when the slab is illuminated by light of wavelength  $\lambda$  from below.



- It is observed that on the painted surface, a very dim dark circle of diameter  $d$  is surrounded by an illuminated region. Explain this observation with an illustrated diagram.
- Derive the value of  $n_s$  in terms of the measured quantities  $d$  and  $t$ .
- Suppose the setup is immersed in a liquid with index of refraction  $n' < n_s$ . Describe (both qualitatively and quantitatively if appropriate) what will be observed when light of wavelength  $\lambda$  is illuminated underneath.
- The phenomenon is caused by **total internal reflection**. The dark circle is the region formed by light whose some proportion escapes the slab. The illuminated region outside the circle is where light has reflected back down and strikes the white paint. The boundary of this circle corresponds to the ray that hits the top surface exactly at the critical angle.



(b) Let  $\theta_c$  be the critical angle at the top interface. By Snell's Law ( $n_s \sin \theta_c = n_{air} \sin 90^\circ$ ),

$$\sin \theta_c = \frac{1}{n_s}$$

From the geometry of the critical ray, the ray travels a horizontal distance  $x$  to reach the top and another  $x$  to return to the bottom. The radius of the dark circle is  $R = d/2 = 2x$ . The horizontal distance  $x$  for a slab of thickness  $t$  is

$$x = t \tan \theta_c$$

Thus, the diameter  $d$  is

$$d = 4t \tan \theta_c \quad \implies \quad \tan \theta_c = \frac{d}{4t}$$

Using the trigonometric identity  $\sin \theta = \frac{\tan \theta}{\sqrt{1 + \tan^2 \theta}}$ :

$$\frac{1}{n_s} = \frac{\frac{d}{4t}}{\sqrt{1 + \left(\frac{d}{4t}\right)^2}} = \frac{\frac{d}{4t}}{\frac{\sqrt{16t^2 + d^2}}{4t}} = \frac{d}{\sqrt{16t^2 + d^2}}$$

Solving for  $n_s$ :

$$n_s = \frac{\sqrt{d^2 + 16t^2}}{d}$$

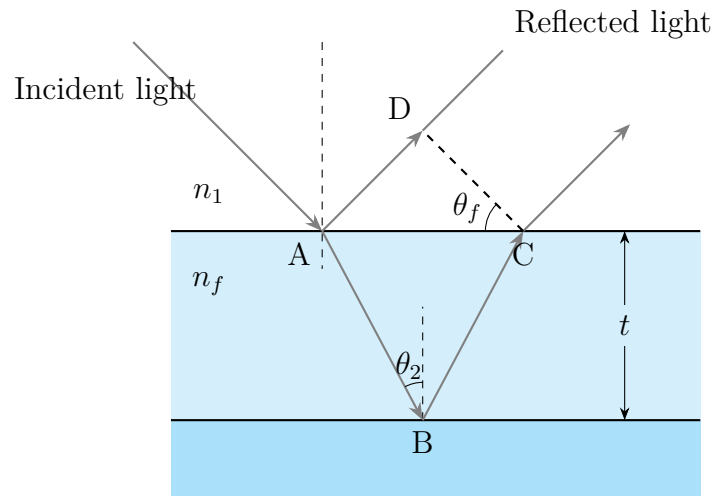
(c) The new critical angle  $\theta'_c$  is determined by

$$\sin(\theta'_c) = \frac{n'}{n_s}$$

As the liquid has a higher refractive index than air ( $n' > 1$ ), the ratio  $\frac{n'}{n_s}$  is larger than  $\frac{1}{n_s}$ . This means  $\sin(\theta'_c)$  increases, and consequently, the critical angle  $\theta'_c$  increases. As  $\tan(\theta)$  increases as the angle increases, the horizontal distance traveled by the rays before they undergo total internal reflection becomes larger, resulting in a larger diameter  $d$  for the dark circle.



## 6.2 Thin-film Interference



Consider a thin film of thickness  $t$  and refractive index  $n_f$ , surrounded by media with refractive indices  $n_1$  (incident side) and  $n_f$  (transmission side). A monochromatic light ray of wavelength  $\lambda_0$  in vacuum is incident at an angle  $\theta_1$ . Inside the film, the ray refracts at an angle  $\theta_f$  given by Snell's law:

$$n_1 \sin \theta_1 = n_f \sin \theta_f$$

Interference occurs between:

- Ray 1: reflected at the top surface of the film,
- Ray 2: transmitted into the film, reflected at the bottom surface, and then transmitted back out.

The two rays generally travel different optical paths and may also undergo phase changes upon reflection. The geometric path difference inside the film is

$$\Delta \ell = 2t \cos \theta_f$$

The corresponding **optical path difference** (OPD) is

$$\Delta = 2n_f t \cos \theta_f$$

This contributes a phase difference

$$\Delta \phi_{\text{path}} = \frac{2\pi}{\lambda_0} (2n_f t \cos \theta_f)$$



Let  $\Delta\phi_{\text{refl}}$  denote the net phase shift due to reflections. The total phase difference between the two reflected rays is

$$\Delta\phi = \frac{4\pi n_f t \cos \theta_f}{\lambda_0} + \Delta\phi_{\text{refl}}$$

For constructive interference,

$$\Delta\phi = 2m\pi, \quad m = 0, 1, 2, \dots$$

For destructive interference,

$$\Delta\phi = (2m + 1)\pi$$

A phase shift of  $\pi$  (equivalent to  $\lambda/2$ ) occurs when light reflects from a boundary leading to a medium of **higher** refractive index. If exactly one of the two reflected rays undergoes a  $\pi$  phase shift, then the conditions become

$$2n_f t \cos \theta_f = \left(m + \frac{1}{2}\right) \lambda_0 \quad (\text{for constructive interference})$$

$$2n_f t \cos \theta_f = m\lambda_0 \quad (\text{for destructive interference})$$

If both of two reflected rays undergo a  $\pi$  phase shift, then the conditions become

$$2n_f t \cos \theta_f = \left(m + \frac{1}{2}\right) \lambda_0 \quad (\text{for destructive interference})$$

$$2n_f t \cos \theta_f = m\lambda_0 \quad (\text{for constructive interference})$$

Also, note that there is no phase shift when light reflects from a boundary leading to a medium of **less** refractive index.

